

Crash insurance and the G10 carry premium: theory notes

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These notes prove the results listed in `theory/README.md`. Results that are standard are marked as such and attributed, and where a published proof follows the same route as the one written here, it is cited. The parts I claim as my own are stated at the start of each section. Throughout, Φ and φ are the standard normal distribution and density functions, $s = \sigma\sqrt{\tau}$, and for forward F and strike K

$$d_{\pm}(K) = \frac{\ln(F/K) \pm s^2/2}{s}.$$

The identity $K\varphi(d_-) = F\varphi(d_+)$ is used repeatedly.

R1. Domestic and foreign risk-neutral measures

This is the change of numeraire of Geman, El Karoui and Rochet (1995); the statement and proof are included for completeness. Girsanov's theorem and Itô's formula are used in the form given by Shreve (2004).

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q}^d)$ satisfy the usual conditions, let B^d, B^f be strictly positive adapted money-market accounts with $B_0^d = B_0^f = 1$, and let $S > 0$ be adapted and càdlàg. Assume $Z_t = S_t B_t^f / (S_0 B_t^d)$ is a true $\mathbb{Q}^d$ -martingale and define $d\mathbb{Q}^f / d\mathbb{Q}^d = Z_T$ on $\mathcal{F}_T$.

Proposition 1. (a) $\mathbb{Q}^f$ is a probability measure equivalent to $\mathbb{Q}^d$. (b) For $\mathcal{F}_T$ -measurable $X \geq 0$ or $X \in L^1(\mathbb{Q}^f)$, $\mathbb{E}^{\mathbb{Q}^f}[X | \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t] / Z_t$. (c) An adapted process Y with $Y_t \in L^1(\mathbb{Q}^f)$ is a $\mathbb{Q}^f$ -martingale if and only if ZY is a $\mathbb{Q}^d$ -martingale; in particular a foreign price V^f satisfies: V^f / B^f is a $\mathbb{Q}^f$ -martingale if and only if $SV^f / B^d = S_0 Z (V^f / B^f)$ is a $\mathbb{Q}^d$ -martingale. (d) If $dS_t / S_t = (r_d - r_f) dt + \sigma dW_t$ under $\mathbb{Q}^d$ with constant coefficients, then $W_t^f = W_t - \sigma t$ is a $\mathbb{Q}^f$ -Brownian motion, $d(1/S_t) = (1/S_t)[(r_f - r_d) dt - \sigma dW_t^f]$, and $\mathbb{E}^{\mathbb{Q}^d} S_T = F$, $\mathbb{E}^{\mathbb{Q}^f}(1/S_T) = 1/F$ with $F = S_0 e^{(r_d - r_f)T}$.

Proof. (a) $Z_T > 0$ and $\mathbb{E}^{\mathbb{Q}^d} Z_T = Z_0 = 1$, so $\mathbb{Q}^f$ is a probability measure; positivity of Z_T gives equivalence. (b) For $\mathcal{F}_t$ -measurable $Y \geq 0$, the tower property and the martingale property give $\mathbb{E}^{\mathbb{Q}^f} Y = \mathbb{E}^{\mathbb{Q}^d}[Z_T Y] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y]$. For $A \in \mathcal{F}_t$,

$$\mathbb{E}^{\mathbb{Q}^f}[X \mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^d}[\mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t] \mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^f}\left[\frac{\mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t]}{Z_t} \mathbf{1}_A\right],$$

which characterises the conditional expectation. (c) Apply (b) on $[s, t]$: $\mathbb{E}^{\mathbb{Q}^f}[Y_t | \mathcal{F}_s] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t | \mathcal{F}_s] / Z_s$, so Y is a $\mathbb{Q}^f$ -martingale exactly when $\mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t | \mathcal{F}_s] = Z_s Y_s$. (d) Here $Z_t = \exp(\sigma W_t - \sigma^2 t / 2)$, so Girsanov's theorem gives the first claim, and Itô's formula for $1/S$ gives the dynamics. The expectations follow from the lognormal law under each measure. $\square$

Remark 2. Jensen's inequality gives $\mathbb{E}^{\mathbb{Q}^d}(1/S_T) > 1/\mathbb{E}^{\mathbb{Q}^d} S_T = 1/F$. There is no contradiction with (d): the foreign investor's forward price is an expectation under $\mathbb{Q}^f$, not $\mathbb{Q}^d$ (Siegel's paradox). If Z is only a strict local martingale, then $\mathbb{E} Z_T < 1$ and Z_T does not define an equivalent probability measure. Carr, Fisher and Ruf (2014) construct the foreign measure in that case as a Föllmer measure that is not equivalent to $\mathbb{Q}^d$, and restore the domestic-foreign symmetry with a modified pricing operator.

R2. Arbitrage-free call prices

The link between the second strike derivative of call prices and state prices is due to Breeden and Litzenberger (1978). The characterisation below is a known result: Föllmer and Schied (2004, Lemma 7.23) show that call prices consistent with a linear, positive pricing rule are the call prices of a unique probability measure with the forward as its mean, by the same argument through the right derivative of the call-price function and Fubini's theorem. I state it in the form used by the arbitrage checks and write out the proof for completeness.

Proposition 3. *Let $c : [0, \infty) \rightarrow \mathbb{R}$ and $F > 0$. There is a random variable $X \geq 0$ with $\mathbb{E}X = F$ and $c(K) = \mathbb{E}(X - K)^+$ for all $K \geq 0$ if and only if (i) c is convex, (ii) $c(0) = F$, (iii) $c'_+(0) \geq -1$ and (iv) $c(K) \rightarrow 0$ as $K \rightarrow \infty$. The law of X is then unique, with distribution function $G = 1 + c'_+$ on $[0, \infty)$ and $\mathbb{P}(X = 0) = 1 + c'_+(0)$.*

Proof. Necessity. Convexity of c follows from convexity of $K \mapsto (x - K)^+$ and linearity of expectation, and $c(0) = \mathbb{E}X = F$. Since $0 \leq (X - K)^+ \leq X \in L^1$ and $(X - K)^+ \rightarrow 0$ pointwise, dominated convergence gives (iv). For $h > 0$, the difference quotient $[(X - K - h)^+ - (X - K)^+]/h$ is bounded by 1 in absolute value and converges to $-\mathbf{1}\{X > K\}$ as $h \downarrow 0$, so $c'_+(K) = -\mathbb{P}(X > K)$. In particular $c'_+(0) = -\mathbb{P}(X > 0) \geq -1$.

Sufficiency. A convex function on $[0, \infty)$ has a right derivative c'_+ that is nondecreasing and right-continuous, and c is absolutely continuous on compact intervals with $c(b) - c(a) = \int_a^b c'_+(u) du$. If $c'_+(K_0) > 0$ for some K_0 , then $c(K) \geq c(K_0) + c'_+(K_0)(K - K_0) \rightarrow \infty$, contradicting (iv); so $c'_+ \leq 0$, c is nonincreasing and, by (iv), $c \geq 0$. Let $L = \lim_{K \rightarrow \infty} c'_+(K) \leq 0$. Since c'_+ is nondecreasing, $c'_+ \leq L$ everywhere, so $c(K) \leq F + LK$; $L < 0$ would force $c < 0$ for large K . Hence $L = 0$.

Define $G(K) = 0$ for $K < 0$ and $G(K) = 1 + c'_+(K)$ for $K \geq 0$. Then G is nondecreasing and right-continuous, $G(0) = 1 + c'_+(0) \geq 0$ by (iii), and $G(K) \rightarrow 1$. It is therefore the distribution function of a random variable $X \geq 0$ (Lebesgue–Stieltjes construction). By Tonelli's theorem,

$$\begin{aligned} \mathbb{E}(X - K)^+ &= \int_K^\infty \mathbb{P}(X > u) du = \int_K^\infty (1 - G(u)) du \\ &= - \int_K^\infty c'_+(u) du = c(K) - \lim_{b \rightarrow \infty} c(b) = c(K), \end{aligned}$$

and $\mathbb{E}X = c(0) = F$. Uniqueness holds because c determines c'_+ and hence G . $\square$

Remark 4. Extending c by $c(K) = F - K$ for $K < 0$, the second distributional derivative of c on $\mathbb{R}$ is the law of X ; the jump of c' at 0 carries the atom $\mathbb{P}(X = 0)$. On a finite strike grid the necessary conditions reduce to slopes in $[-1, 0]$ that are nondecreasing, which is the discrete test used in `src/qef/fx/arbitrage.py`. The implementation also applies the density condition on implied total variance of Gatheral and Jacquier (2014, eq. (2.1) and Lemma 2.2).

R3. Spanning

The representation of a payoff by bonds, forwards and out-of-the-money options is the spanning formula of Carr and Madan (2001; first published 1998). The version below states explicit sufficient hypotheses, including the integrability condition under which expectations can be taken term by term. The proof is the standard Taylor expansion with integral remainder, written out under those hypotheses. In R3 and R4, X is the terminal exchange rate and $\mathbb{E}$ and $\mathbb{Q}$ refer to the same pricing measure.

Proposition 5. *Let $f : (0, \infty) \rightarrow \mathbb{R}$ have a locally absolutely continuous derivative f' , so that f'' exists almost everywhere and is locally integrable. Let $X > 0$ almost surely with $\mathbb{E}X = F < \infty$,*

and write $P(K) = \mathbb{E}(K - X)^+$ and $C(K) = \mathbb{E}(X - K)^+$. If

$$I = \int_0^F |f''(K)| P(K) dK + \int_F^\infty |f''(K)| C(K) dK < \infty,$$

then $f(X) \in L^1$ and

$$\mathbb{E}f(X) = f(F) + \int_0^F f''(K) P(K) dK + \int_F^\infty f''(K) C(K) dK.$$

Proof. Fix $x > 0$. Absolute continuity of f and f' on the interval between x and F gives $f(x) - f(F) = \int_F^x f'(u) du$ and $f'(u) = f'(F) + \int_F^u f''(K) dK$. Substituting and applying Fubini's theorem on the bounded interval (where f'' is integrable),

$$f(x) = f(F) + f'(F)(x - F) + \int_F^x f''(K)(x - K) dK.$$

For $x > F$ the last integral equals $\int_F^\infty f''(K)(x - K)^+ dK$; for $x < F$ it equals $\int_x^F f''(K)(K - x) dK = \int_0^F f''(K)(K - x)^+ dK$. Hence, for every $x > 0$,

$$f(x) = f(F) + f'(F)(x - F) + \int_0^F f''(K)(K - x)^+ dK + \int_F^\infty f''(K)(x - K)^+ dK.$$

The map $(K, \omega) \mapsto f''(K)(K - X(\omega))^+$ is jointly measurable. By Tonelli's theorem, $\mathbb{E} \int_0^F |f''(K)|(K - X)^+ dK = \int_0^F |f''(K)| P(K) dK$, and similarly for the call term, so both are at most $I < \infty$. Each term on the right is therefore integrable, hence so is $f(X)$, and Fubini's theorem allows expectation and integral to be exchanged. Since $\mathbb{E}(X - F) = 0$, the formula follows. $\square$

Remark 6. For $f(x) = \ln(x/F)$, $f''(K) = -K^{-2}$, which gives the log contract $\mathbb{E} \ln(X/F) = -\int_0^F P(K) K^{-2} dK - \int_F^\infty C(K) K^{-2} dK$. For $f(x) = (\ln(x/F))^2$, $f''(K) = 2(1 + \ln(F/K))/K^2$, which changes sign at $K = eF$; the absolute-value condition in the proposition covers such weights. The moment contracts of Bakshi, Kapadia and Madan (2003, Theorem 1) are of this type; they expand around the spot price with a constant interest rate, whereas the contracts here are centred at the forward F . Quotes observe P and C only between the 10Δ strikes, so the tails of these integrals must be bounded (R4).

R4. Bounds on the unobserved tails

Jiang and Tian (2005, Proposition 2 and the Appendix) bound the truncation error of model-free implied variance beyond the quoted strike range. The bounds below serve the same purpose under a different assumption, a power-type monotonicity of the distribution function in each tail, which makes them computable from the last quoted prices and extends them to the weights of the higher moment contracts. Lee (2004) relates the number of finite moments of the underlying to the limiting slope of implied variance in each wing, the asymptotic counterpart of the tail exponents used here. The lemma, the proposition and their proofs are my own; I have not found them in this form in the sources I consulted.

Let $G(u) = \mathbb{Q}(X \leq u)$ and $\bar{G}(u) = \mathbb{Q}(X > u)$. By Tonelli's theorem, $P(K) = \int_0^K G(u) du$ and $C(K) = \int_K^\infty \bar{G}(u) du$. Let $K_{\min} < F < K_{\max}$ be the lowest and highest strikes at which prices are observed.

Lemma 7. (a) If $u \mapsto G(u)/u^\gamma$ is nondecreasing on $(0, K_{\min}]$ for some $\gamma > 0$, then $P(K) \leq P(K_{\min})(K/K_{\min})^{\gamma+1}$ for $0 < K \leq K_{\min}$. (b) If $u \mapsto u^\eta \bar{G}(u)$ is nonincreasing on $[K_{\max}, \infty)$ for some $\eta > 1$, then $C(K) \leq C(K_{\max})(K/K_{\max})^{1-\eta}$ for $K \geq K_{\max}$.

Proof. (a) With $\lambda = K/K_{\min} \in (0, 1]$ and the substitution $u = \lambda v$, $P(K) = \lambda \int_0^{K_{\min}} G(\lambda v) dv$. Since $\lambda v \leq v$ and $G(w)/w^\gamma$ is nondecreasing, $G(\lambda v) \leq \lambda^\gamma G(v)$. Hence $P(K) \leq \lambda^{\gamma+1} \int_0^{K_{\min}} G(v) dv = \lambda^{\gamma+1} P(K_{\min})$. (b) With $\lambda = K/K_{\max} \geq 1$ and $u = \lambda v$, $C(K) = \lambda \int_{K_{\max}}^\infty \bar{G}(\lambda v) dv$, and $\bar{G}(\lambda v) \leq \lambda^{-\eta} \bar{G}(v)$ because $w^\eta \bar{G}(w)$ is nonincreasing. Hence $C(K) \leq \lambda^{1-\eta} C(K_{\max})$. $\square$

Proposition 8. *Under the hypotheses of the lemma, for any measurable $w \geq 0$ on $(0, K_{\min}]$,*

$$\int_0^{K_{\min}} w(K) P(K) dK \leq \frac{P(K_{\min})}{K_{\min}^{\gamma+1}} \int_0^{K_{\min}} w(K) K^{\gamma+1} dK.$$

In particular $\int_0^{K_{\min}} P(K) K^{-2} dK \leq P(K_{\min})/(\gamma K_{\min})$. For $w(K) = 2(1 + \ln(F/K))K^{-2}$,

$$\int_0^{K_{\min}} w(K) P(K) dK \leq \frac{2P(K_{\min})}{K_{\min}} \left[\frac{1 + \ln(F/K_{\min})}{\gamma} + \frac{1}{\gamma^2} \right].$$

For the upper tail and any measurable w on $[K_{\max}, \infty)$,

$$\int_{K_{\max}}^\infty |w(K)| C(K) dK \leq C(K_{\max}) K_{\max}^{\eta-1} \int_{K_{\max}}^\infty |w(K)| K^{1-\eta} dK;$$

in particular $\int_{K_{\max}}^\infty C(K) K^{-2} dK \leq C(K_{\max})/(\eta K_{\max})$, and for $w(K) = 2(1 - \ln(K/F))K^{-2}$ with $F < K_{\max} \leq eF$ and $\ell = \ln(K_{\max}/F)$,

$$\int_{K_{\max}}^\infty |w(K)| C(K) dK \leq \frac{2C(K_{\max})}{K_{\max}} \left[\frac{1-\ell}{\eta} - \frac{1}{\eta^2} + \frac{2e^{-\eta(1-\ell)}}{\eta^2} \right].$$

Proof. The first inequality is the lemma multiplied by $w \geq 0$ and integrated. For $w(K) = K^{-2}$, $\int_0^{K_{\min}} K^{\gamma-1} dK = K_{\min}^\gamma/\gamma$. For the logarithmic weight, integration by parts gives $\int_0^{K_{\min}} K^{\gamma-1} \ln(F/K) dK = K_{\min}^\gamma \ln(F/K_{\min})/\gamma + K_{\min}^\gamma/\gamma^2$, the boundary term at zero vanishing because $K^\gamma \ln(F/K) \rightarrow 0$. The upper-tail bounds follow from part (b) of the lemma. For $w(K) = K^{-2}$, $\int_{K_{\max}}^\infty K^{-1-\eta} dK = K_{\max}^{-\eta}/\eta$. For the logarithmic weight, the substitution $u = \ln(K/F)$ gives $F^{-\eta} \int_\ell^\infty |1-u| e^{-\eta u} du$, which is split at $u = 1$ and integrated by parts. $\square$

Remark 9. The bounds use only observed wing prices and an assumed tail exponent. A tail term whose weight has constant sign lies between 0 and its bound, taken with the sign of the weight; a term with a sign-changing weight lies in $[-B, B]$, where B bounds the integral of $|w|$. Because γ and η are not observed, risk-neutral variance and skewness are reported as intervals over a stated range of exponents, obtained by propagating the intervals of the individual contracts, never as point values. Between $K_{\min}$ and $K_{\max}$, P and C are taken from the calibrated smile and the integrals are to be computed by composite quadrature (research design, section 6), with F a node and each side integrated separately, because the out-of-the-money integrand has a kink at F where $P' - C' = 1$. With node spacing h on a side $[a, b]$ and an even number of subintervals for Simpson's rule, the composite trapezium error is at most $(b-a)h^2 \max |g''|/12$ if $g \in C^2[a, b]$, and the composite Simpson error at most $(b-a)h^4 \max |g^{(4)}|/180$ if $g \in C^4[a, b]$ (Süli and Mayers, 2003).

R5(b). Premium-adjusted call delta

The premium-adjusted delta, its non-monotonicity in the strike and the market practice of taking the strike to the right of the delta maximum are described by Reiswich and Wystup (2012; working-paper version of 20 March 2010, Sections 1.3 and 1.5), who also show that the premium-adjusted delta lies below the simple delta because the call price is positive, and state

that the maximum solves $\sigma\sqrt{\tau}\Phi(d_-) = \varphi(d_-)$. Jäckel (2020, Section 2, eq. (20)) shows that the premium-adjusted forward delta of a call has a single maximum, located where the inverse Mills ratio equals the total volatility, and takes the larger of the two solutions, which is the strike on the conventional branch. The proposition below states this shape in the notation used here; its proof is the same inverse Mills ratio argument, written out for completeness, and none of it is my own.

Let $D \in (0, 1]$ and $h(K) = D(K/F)\Phi(d_-(K))$ for $K > 0$. With spot delta, D is the base-currency discount factor; with forward delta, $D = 1$.

Lemma 10. *The inverse Mills ratio $\lambda(d) = \varphi(d)/\Phi(d)$ is strictly decreasing on $\mathbb{R}$, with $\lambda(d) \rightarrow \infty$ as $d \rightarrow -\infty$ and $\lambda(d) \rightarrow 0$ as $d \rightarrow \infty$.*

Proof. Differentiation gives $\lambda'(d) = -\lambda(d)[d + \lambda(d)]$, so it suffices that $d\Phi(d) + \varphi(d) > 0$. This function has derivative $\Phi(d) > 0$ and tends to 0 as $d \rightarrow -\infty$; hence $d\Phi(d) + \varphi(d) = \int_{-\infty}^d \Phi(u) du > 0$. The limits follow from $\Phi \rightarrow 0$, $\varphi \rightarrow 0$ and the Mills-ratio asymptotics $\Phi(d) \sim \varphi(d)/|d|$ as $d \rightarrow -\infty$. The lemma is known: the normal density is log-concave, so Φ is log-concave (Bagnoli and Bergstrom, 2005, Section 2.1, Theorem 1) and $\lambda = (\log \Phi)'$ is decreasing, and inequalities of this type for the Mills ratio are classical (Sampford, 1953). The short direct proof above is written out for completeness. $\square$

Proposition 11. *The function h is continuous on $(0, \infty)$, tends to 0 as $K \downarrow 0$ and as $K \rightarrow \infty$, and satisfies $h < D$. There is a unique K^* such that h is strictly increasing on $(0, K^*]$ and strictly decreasing on $[K^*, \infty)$; $K^* = F \exp(-d^*s - s^2/2)$, where d^* is the unique solution of $\lambda(d) = s$. Each $\Delta \in (0, h(K^*))$ has exactly two preimages, one on each side of K^* .*

Proof. As $K \downarrow 0$, $d_- \rightarrow \infty$, so $h \rightarrow 0$. As $K \rightarrow \infty$, $d_- \rightarrow -\infty$ and, by the Mills-ratio asymptotics and the identity above, $K\Phi(d_-) \sim K\varphi(d_-)/|d_-| = F\varphi(d_+)/|d_-| \rightarrow 0$. Moreover $F\Phi(d_+) - K\Phi(d_-)$ is the undiscounted call price, which is positive, so $h < D\Phi(d_+) < D$.

Since $d'_-(K) = -1/(Ks)$,

$$h'(K) = \frac{D}{F} \left[\Phi(d_-) - \frac{\varphi(d_-)}{s} \right] = \frac{D\Phi(d_-)}{Fs} [s - \lambda(d_-)].$$

By the lemma, $s - \lambda(d)$ is strictly increasing in d and changes sign exactly once, at d^* . Because d_- is strictly decreasing in K , $h' > 0$ for $K < K^*$ and $h' < 0$ for $K > K^*$, where $d_-(K^*) = d^*$. The stated formula for K^* inverts d_- . The final claim follows from the intermediate value theorem and strict monotonicity on each side. $\square$

Remark 12. The market convention selects the preimage in $[K^*, \infty)$. At one month and 25 Δ this root lies far to the right of K^* ; bracketing matters at long maturities and high volatilities.

R5(c). Delta-to-strike inversion with a smile

Reiswich (2010, Section 3.8 and Fig. 3.6) shows that the delta implied by a smile need not be monotone in the strike, so that a quoted delta can correspond to several strikes or to none. Fukasawa (2012, Lemma 2.1 and Propositions 2.1–2.2) shows, by the same derivative calculation as below, that d_+ and d_- are strictly decreasing in the strike for every arbitrage-free smile, and Mingone (2023, Lemma 4.1) links this monotonicity to a unique strike for each delta. The part that is my own is only the explicit slope bound in the proposition, which can be checked on a fitted smile that is not assumed to be free of arbitrage, stated here for pips spot and forward deltas with the discount factor D .

Proposition 13. Let $\sigma : (0, \infty) \rightarrow (0, \infty)$ be continuously differentiable and let $\Delta_\phi(K) = \phi D \Phi(\phi d_+(K, \sigma(K)))$ be the pips spot delta (D the base-currency discount factor) or forward delta ($D = 1$) of a call ($\phi = 1$) or put ($\phi = -1$). On an interval J , if

$$|\sigma'(K)| K \sqrt{\tau} |d_-(K, \sigma(K))| < 1 \quad \text{for all } K \in J,$$

then Δ_ϕ is strictly decreasing on J , and each delta attained on J is attained at exactly one strike in J .

Proof. With $d_+ = \ln(F/K)/(\sigma\sqrt{\tau}) + \sigma\sqrt{\tau}/2$, one has $\partial_K d_+ = -1/(K\sigma\sqrt{\tau})$ and $\partial_\sigma d_+ = -d_-/\sigma$. By the chain rule,

$$\frac{d}{dK} d_+(K, \sigma(K)) = -\frac{1}{K\sigma\sqrt{\tau}} \left[1 + K\sqrt{\tau} d_- \sigma'(K) \right],$$

which is negative on J under the stated condition. Since $\Delta'_\phi(K) = \phi^2 D \varphi(d_+) \frac{d}{dK} d_+ = D \varphi(d_+) \frac{d}{dK} d_+$, the function Δ_ϕ is strictly decreasing on J , and injectivity gives uniqueness. $\square$

Remark 14. In log-moneyness $k = \ln(K/F)$, $K\sigma'(K) = \partial\sigma/\partial k$, so the condition reads $|\partial\sigma/\partial k| \sqrt{\tau} |d_-| < 1$. Across the calibrated one-month smiles of the primary sample, its maximum over ± 4 ATM standard deviations is 0.76, so the smile delta-to-strike map is injective on that range at every month-end. For premium-adjusted delta $\phi D(K/F) \Phi(\phi d_-)$ the analogous derivative is $\phi(D/F) [\Phi(\phi d_-) + \phi K \varphi(d_-) \frac{d}{dK} d_-]$ with $\frac{d}{dK} d_- = -[1 + K\sqrt{\tau} d_+ \sigma'(K)]/(K\sigma\sqrt{\tau})$; numerically, the premium-adjusted put delta is strictly monotone and the call delta strictly decreasing to the right of its maximum on the same range at every month-end.

R6. Diffusive null of the hedge cost

The comparison of hedged and unhedged carry returns follows Jurek (2014), who reads the resulting ratio as an upper bound on the crash share and notes that an unlevered option hedge also gives up part of the diffusive premium (working-paper version of August 2013, Section 2.2). Farhi et al. (2015, Section 5.1, eqs. (8)–(9)) state the leading-order version of the result below: a put of delta Δ_P leaves a fraction $1 + \Delta_P$ of the Gaussian risk premium. The mean-value form and the error bound below are my own short derivation.

Let the undiscounted put value with lognormal forward G and volatility σ be

$$p(G) = K \Phi(-d_-(G)) - G \Phi(-d_+(G)), \quad d_\pm(G) = \frac{\ln(G/K) \pm s^2/2}{s}.$$

Using $K\varphi(d_-) = G\varphi(d_+)$, $p'(G) = -\Phi(-d_+(G))$.

Suppose that under $\mathbb{Q}$, S_τ is lognormal with mean F and log-variance s^2 , and that under $\mathbb{P}$ it is lognormal with the same log-variance and mean $Fe^{\lambda\tau}$ (no jumps, no volatility premium). A unit long forward costs nothing and has $\mathbb{P}$ -expected payoff $\mu_U = F(e^{\lambda\tau} - 1)$. Adding a put bought at its $\mathbb{Q}$ -value, with the premium financed to τ , gives $\mu_H = \mu_U + p(Fe^{\lambda\tau}) - p(F)$.

Proposition 15. If $\lambda \neq 0$, the hedge-cost ratio $\theta_0 = 1 - \mu_H/\mu_U$ satisfies $\theta_0 = \Phi(-d_+(\xi))$ for some ξ between F and $Fe^{\lambda\tau}$, and

$$|\theta_0 - \Phi(-d_1)| \leq \varphi(0) \frac{|\lambda|\sqrt{\tau}}{\sigma}, \quad d_1 = d_+(F).$$

Proof. By the mean value theorem, $p(Fe^{\lambda\tau}) - p(F) = -\Phi(-d_+(\xi)) F(e^{\lambda\tau} - 1)$ for some ξ between the two forwards, so $\theta_0 = -[p(Fe^{\lambda\tau}) - p(F)]/\mu_U = \Phi(-d_+(\xi))$. Since $|d_+(\xi) - d_+(F)| = |\ln(\xi/F)|/s \leq |\lambda|\tau/s$ and Φ is Lipschitz with constant $\varphi(0)$, the bound follows. $\square$

Remark 16. With no crash premium at all, hedging with a put of forward delta $-\Phi(-d_1)$ surrenders approximately that fraction of the carry premium: about 0.25 for a 25Δ put and 0.10 for a 10Δ put. This is the null against which the realised ratio θ_{UB} and the skew component of E3 are read.

R7. Entropy decomposition of currency premia

The decomposition is due to Backus, Foresi and Telmer (2001): with the exchange rate equal to the ratio of the two pricing kernels (their Proposition 1), the expected excess return on a forward position is the difference between the log of the expectation and the expectation of the log of the two kernels (their eq. (11)), and its higher cumulants enter only through that difference (eqs. (12)–(13)). Conditional entropy, as this difference is now called, and the entropy bound are as in Backus, Chernov and Zin (2014), who relate the bound to Bansal and Lehmann (1997) and Alvarez and Jermann (2005). Lustig and Verdelhan (2019, Section 2.5) restate the decomposition with a wedge between the exchange rate and the ratio of kernels, and Jiang, Krishnamurthy and Lustig (2021) link the value of the dollar to a convenience yield inferred from the Treasury basis, a covered-parity deviation measured with government bond yields. The identity below writes the same decomposition with the covered-interest-parity basis as the wedge on the forward side; the derivation is elementary.

Fix t and write $\mathbb{E}_t$ for $\mathbb{E}[\cdot \mid \mathcal{F}_t]$. For a positive random variable M with $\mathbb{E}_t|\log M| < \infty$, the conditional entropy is $L_t(M) = \log \mathbb{E}_t M - \mathbb{E}_t \log M \geq 0$. Let $M, M^* > 0$ be $\mathcal{F}_{t+1}$ -measurable discount factors in L^2 that price domestic and foreign payoffs in their own currencies, with $\mathbb{E}_t M = e^{-r_t}$, $\mathbb{E}_t M^* = e^{-r_t^*}$, $\mathbb{E}_t|\log M|, \mathbb{E}_t|\log M^*| < \infty$ and $MS_{t+1}/S_t \in L^2$, where S is the domestic price of foreign currency. Write $m = \log M$, $\Delta s = \log(S_{t+1}/S_t)$, and $x_t = (f_t - s_t) - (r_t - r_t^*)$ for the covered-interest-parity basis. With exchange rates quoted in foreign currency per USD, as in Du, Tepper and Verdelhan (2018; NBER Working Paper 23170, eq. (4)), x_t is minus their basis.

Lemma 17. *Suppose every foreign payoff $X^* \in L^2(\mathcal{F}_{t+1})$ is traded and priced consistently in both currencies, $S_t \mathbb{E}_t[M^* X^*] = \mathbb{E}_t[MS_{t+1} X^*]$. Then $M^* = MS_{t+1}/S_t$ almost surely.*

Proof. The difference $D = M^* - MS_{t+1}/S_t$ lies in L^2 and satisfies $\mathbb{E}_t[DX^*] = 0$ for all $X^* \in L^2$. Taking $X^* = D$ gives $\mathbb{E}_t D^2 = 0$. $\square$

Proposition 18. *Under the lemma's conclusion, (i) $\mathbb{E}_t[\Delta s + r_t^* - r_t] = L_t(M) - L_t(M^*)$ and $\mathbb{E}_t[s_{t+1} - f_t] = L_t(M) - L_t(M^*) - x_t$. (ii) If $k(u) = \log \mathbb{E}_t e^{um}$ is finite near 0 and its Maclaurin series has radius of convergence greater than 1, then $L_t(M) = \sum_{j \geq 2} \kappa_{j,t}/j!$, where $\kappa_{j,t}$ are the conditional cumulants of m . (iii) For any gross return $R > 0$ with $\mathbb{E}_t[MR] = 1$ and $\mathbb{E}_t|\log R| < \infty$, $L_t(M) \geq \mathbb{E}_t \log R - r_t$.*

Proof. (i) By definition, $\mathbb{E}_t m = -r_t - L_t(M)$ and $\mathbb{E}_t m^* = -r_t^* - L_t(M^*)$. The lemma gives $\Delta s = m^* - m$, so $\mathbb{E}_t \Delta s = r_t - r_t^* + L_t(M) - L_t(M^*)$. The second identity follows from $s_{t+1} - f_t = \Delta s - (f_t - s_t) = \Delta s - (r_t - r_t^*) - x_t$. (ii) $L_t(M) = k(1) - k'(0)$ with $k'(0) = \kappa_{1,t}$, and $k(1) = \sum_{j \geq 1} \kappa_{j,t}/j!$ under the stated convergence. (iii) By Jensen's inequality, $0 = \log \mathbb{E}_t[MR] \geq \mathbb{E}_t m + \mathbb{E}_t \log R$, so $L_t(M) = -r_t - \mathbb{E}_t m \geq \mathbb{E}_t \log R - r_t$. $\square$

Remark 19. The identity in (i) needs only that some pair of discount factors prices bonds and exchange rates consistently; completeness makes the pair unique. Brandt, Cochrane and Santa-Clara (2006) show that the smoothness of exchange rates relative to the volatility of marginal utility implied by equity premia requires a very high degree of international risk sharing. Options on S reveal the risk-neutral law of $\Delta s = m^* - m$, not the separate cumulants of m and m^* , so they constrain but do not identify the split of $L_t(M) - L_t(M^*)$ across cumulant orders. Hedged and unhedged carry returns give two lower bounds on the same $L_t(M)$ through (iii).

R8. Local well-posedness of the smile calibration

Reiswich (2010, Theorems 2 and 3 and Appendices A and B) uses the implicit function theorem to show that a volatility-strike function exists near the at-the-money point and that the market-strangle calibration of the simplified parabolic smile is locally solvable for the smile strangle

near a zero risk reversal. The proposition below applies the same argument to the three SABR parameters and the three quotes. Le Floch and Kennedy (2014, Section 4, eqs. (30)–(31)) invert the leading-order expansion of the Hagan formula (Hagan et al., 2002, eq. (3.1a)) in closed form, recovering (α, ρ, ν) from the at-the-money level, skew and curvature, which is an approximate, global counterpart of the local and exact statement here. The three calibration equations are the conditions of Reiswich and Wystup (2012; working-paper version, Section 2, eqs. (17), (18) and (25)), to which Reiswich (2010, Section 3.2) calibrates SABR by least squares, and Henrard (2011) uses the implicit function theorem in the same way for any calibration with as many instruments as parameters. In the lemma, the value of $z/x(z)$ at $z = 0$ and its expansion are known (Choi and Wu, 2021, eqs. (4)–(5)), and the integral representation is Hadamard’s lemma (Milnor, 1963, Section 2, Lemma 2.1). What I claim as my own is the explicit statement that the $\beta = 1$ formula is jointly smooth on the stated domain, including $\nu = 0$, and the local well-posedness statement for SABR under conditions (i) to (iv), written for FX delta conventions; I have not found these in this form in the sources I consulted.

Lemma 20. *For $\beta = 1$ the Hagan, Kumar, Lesniewski and Woodward (2002) implied volatility $\sigma(K; \theta)$, $\theta = (\alpha, \rho, \nu)$, is infinitely differentiable on $\{K > 0\} \times (0, \infty) \times (-1, 1) \times [0, \infty)$.*

Proof. For $\beta = 1$, $\sigma = \alpha (z/x(z)) [1 + (\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24)\tau]$ with $z = (\nu/\alpha) \ln(F/K)$ and $x(z) = \ln[(\sqrt{1 - 2\rho z + z^2} + z - \rho)/(1 - \rho)]$. Since $(1 - 2\rho z + z^2) - (z - \rho)^2 = 1 - \rho^2 > 0$, the square root exceeds $|z - \rho|$, so the argument of the logarithm is positive and x is smooth. Moreover $x'(z) = (1 - 2\rho z + z^2)^{-1/2} > 0$ and $x(0) = 0$, so $x(z, \rho) = z h(z, \rho)$ with $h(z, \rho) = \int_0^1 (1 - 2\rho t z + t^2 z^2)^{-1/2} dt$. Since $1 - 2\rho t z + t^2 z^2 \geq 1 - \rho^2 > 0$, the integrand is smooth and positive on $\mathbb{R} \times (-1, 1) \times [0, 1]$, so by differentiation under the integral h is smooth and positive, and $z/x(z) = 1/h(z, \rho)$ (equal to 1 at $z = 0$) is jointly smooth in (z, ρ) . Compositions and products of smooth maps are smooth. $\square$

Fix F, τ, D and the delta convention, and write $q = (\sigma_{ATM}, RR, BF)$ for the quotes. For parameters θ , let $K_A(\theta)$ solve $\ln(K/F) \mp \sigma(K; \theta)^2 \tau / 2 = 0$ (the delta-neutral-straddle strike, with the sign given by the premium convention). Let $K_c(\theta)$ and $K_p(\theta)$ solve $\Delta(K, \sigma(K; \theta)) = \pm 0.25$. Let $K_c^{ms}(q)$ and $K_p^{ms}(q)$ be the $\pm 25\Delta$ strikes at the single volatility $\sigma_{ATM} + BF$. Define $\Psi = (\Psi_1, \Psi_2, \Psi_3)$ by

$$\begin{aligned} \Psi_1 &= \sigma(K_A; \theta) - \sigma_{ATM}, & \Psi_2 &= \sigma(K_c; \theta) - \sigma(K_p; \theta) - RR, \\ \Psi_3 &= \sum_{i \in \{c, p\}} [V(K_i^{ms}, \sigma(K_i^{ms}; \theta)) - V(K_i^{ms}, \sigma_{ATM} + BF)], \end{aligned}$$

where V is the undiscounted Garman–Kohlhagen premium of the corresponding option. The calibrated smile solves $\Psi(\theta, q) = 0$. The market-strangle condition is that of Reiswich and Wystup (2012). For θ near θ^* , $K_A(\theta)$, $K_c(\theta)$ and $K_p(\theta)$ denote the C^1 branches through the strikes at θ^* given by the implicit function theorem under (i) and (ii) below; for a premium-adjusted call, the strike at θ^* is the root with $\partial_K \Delta < 0$ (the conventional branch), which is the root that `strike_from_delta_smile` selects.

Proposition 21. *Let $\Psi(\theta^*, q^*) = 0$. Suppose that at (θ^*, q^*) : (i) $1 \mp K\sigma\sigma'(K)\tau \neq 0$ at K_A ; (ii) the strike derivative of $\Delta(K, \sigma(K; \theta^*))$ is non-zero at K_c and K_p (for pips delta, for instance under the condition of R5(c); for premium-adjusted delta the derivative in the remark after R5(c) must be checked directly, since a flat smile satisfies the R5(c) condition while the call derivative vanishes at K^*); (iii) for premium-adjusted delta, the 25Δ call strike at the flat volatility $\sigma_{ATM}^* + BF^*$ lies strictly to the right of the maximiser, $K_c^{ms} > K^*$ (equivalently $0.25 < h(K^*)$ in the notation of R5(b)); and (iv) $D_\theta \Psi(\theta^*, q^*)$ is non-singular. Then there are neighbourhoods $U \ni q^*$ and $W \ni \theta^*$ and a unique C^1 map $\theta : U \rightarrow W$ with $\Psi(\theta(q), q) = 0$. In particular θ^* is the only calibration in W with strikes on these branches, and the calibrated smile varies smoothly with the quotes.*

Proof. Since x has the sign of z , $z/x(z) > 0$, so $\sigma(\cdot; \theta) > 0$ for all K exactly when $1 + (\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24)\tau > 0$. This holds at θ^* because $\sigma(K_A; \theta^*) = \sigma_{ATM} > 0$, hence near θ^* , and Δ and V are evaluated only at positive volatilities. By the lemma, each defining equation of K_A , K_c , K_p is then C^1 in (K, θ) . Condition (i) is the non-vanishing of the K -derivative of the first; condition (ii) that of the other two. The implicit function theorem therefore makes K_A , K_c and K_p C^1 functions of θ near θ^* . The strikes K_i^{ms} depend on q through $\sigma_{ATM} + BF$: in closed form for pips delta, $K_{c,p}^{ms} = F \exp(\frac{1}{2}s_q^2 \mp s_q \Phi^{-1}(0.25/D))$ with $s_q = (\sigma_{ATM} + BF)\sqrt{\tau}$; and by the implicit function theorem for premium-adjusted delta, since $h' < 0$ on (K^*, ∞) by the proof of R5(b) and condition (iii), and $\partial_K[-D(K/F)\Phi(-d_-)] = -(D/F)[\Phi(-d_-) + \varphi(d_-)/s] < 0$ for the put. Hence G is C^1 in (θ, q) near (θ^*, q^*) , and the implicit function theorem applied to G with (iv) gives the claim. $\square$

Remark 22. Condition (iv) is checked at every calibration through the finite-difference Jacobian J returned by the solver. J is taken in $x = (\ln \alpha, \operatorname{atanh} \rho, \ln \nu)$, with Ψ_3 divided by the flat-volatility strangle vega $\mathcal{V}$, so $J = \operatorname{diag}(1, 1, \mathcal{V}^{-1}) D_\theta \Psi \operatorname{diag}(\alpha, 1 - \rho^2, \nu)$, which is non-singular exactly when $D_\theta \Psi$ is. Across the primary sample $\operatorname{cond}(J)$ is at most 125 under the market-strangle reading and at most 634 under the smile-strangle reading. Under the smile-strangle reading, Ψ_3 is replaced by $\frac{1}{2}[\sigma(K_c; \theta) + \sigma(K_p; \theta)] - \sigma_{ATM} - BF$, and the proposition and its proof hold without condition (iii). The proposition is local: it does not assert existence of a solution for arbitrary quotes, and it does not exclude other solutions far from θ^* . The implementation therefore restarts from several points and records non-convergence.

R10. Admissible tail exponents and sharp identification of moments

R4 bounds each moment contract separately, and it leaves open which exponents are compatible with the quotes. This section answers both questions under the hypotheses of R4. The notation is that of R3 and R4, with X the terminal exchange rate under the quote-currency forward measure, $P_0 = P(K_{\min})$, $G_0 = G(K_{\min}) = P'(K_{\min}+)$, $C_0 = C(K_{\max})$ and $\bar{G}_0 = \bar{G}(K_{\max}) = -C'(K_{\max}+)$, all taken from the calibrated smile on $[K_{\min}, K_{\max}]$. Write $H_L(\gamma)$ for the hypothesis that $G(u)/u^\gamma$ is nondecreasing on $(0, K_{\min}]$ and $H_U(\eta)$, $\eta > 1$, for the hypothesis that $u^\eta \bar{G}(u)$ is nonincreasing on $[K_{\max}, \infty)$.

Several ingredients are known, and I credit them where they are used. $H_L(\gamma)$ is the restriction to the lower tail of the reversed hazard rate order, equivalently the star order, of X against the power-function law, and $H_U(\eta)$ is the corresponding statement against the Pareto law (Lando, Arab and Oliveira, 2022, Definition 2); with a density, $H_L(\gamma)$ says that the elasticity $ug(u)/G(u)$ of the distribution function, studied by Veres-Ferrer and Pavia (2022, Sections 2–5), is at least γ in the tail. The quantity $K_{\min}P'(K_{\min})/P(K_{\min})$ is proposed as the natural exponent of a power-law put extrapolation by Benaim, Dodgson and Kainth (2008, footnote 6 of the version consulted), and Qin, Yang, Che and Feng (2026, Propositions 1 and 2) construct the pure power-law tails that match the boundary price and slope. Taleb, Yacrk, Mann, Delic and Spitznagel (2023, Section IV) observe that convexity at the strike where a Pareto tail is attached restricts its index, and Bollerslev, Todorov and Xu (2015, eq. (3.3)) estimate tail decay from log-log slopes of out-of-the-money prices. The reduction of bounds over a class generated by a family of distributions to a moment problem over the mixing measure follows Popescu (2005, Section 3); that extreme points of the resulting set are mixtures of at most two generators is Theorem 2.1 of Winkler (1988), as quoted by Pinelis (2012, Theorem 12); and the description of the bounds as sections of the convex hull of a curve is that of Kemperman (1968, Section 0 and Theorem 1). The breakdown point in (e) below is the identification breakdown point of Horowitz and Manski (1995), generalised by Stoye (2005, 2010), as set out by Masten and Poirier (2020, pp. 42 and 53), and it is defined, as in Kline and Santos (2013, Example 2.3), as the level of

an assumption at which the sign of a parameter stops being identified. Zhuang (2026) studies sharp bounds on option-implied tail quantities from finitely many quotes without a tail-shape hypothesis, and notes that the boundary slope G_0 is itself only set-identified from finite quotes; here it is taken from the calibrated smile.

What I claim as my own is the following: the characterisation of the exponents compatible with the observed boundary price and slope under H_L and H_U , and the uniqueness of the tail within the whole class at the largest exponent; the generating family of the class with the boundary value fixed, the resulting description of the identified set of the moment contracts, and its consequences for R4; the check of setting (i) in (d); and the use of the breakdown point for the sign of option-implied skewness.

Lemma 23. *Let $\gamma > 0$. A nondecreasing right-continuous function G on $(0, K_{\min}]$ with $G(K_{\min}) = G_0$ satisfies $H_L(\gamma)$ if and only if there is a probability measure μ on $[0, 1]$ with*

$$G(K_{\min}z) = G_0 z^\gamma \mu([0, z]), \quad 0 < z < 1.$$

Then, for measurable ϕ with $\int_0^{K_{\min}} G|\phi| du < \infty$,

$$\int_0^{K_{\min}} G(u)\phi(u) du = G_0 K_{\min} \int_{[0,1]} \int_{z_c}^1 z^\gamma \phi(K_{\min}z) dz \mu(dz_c).$$

Proof. Under $H_L(\gamma)$, $\bar{h}(z) = G(K_{\min}z)/(G_0 z^\gamma)$ is nondecreasing and right-continuous on $(0, 1]$ with $0 \leq \bar{h} \leq \bar{h}(1) = 1$. Let μ be the measure with $\mu([0, z]) = \bar{h}(z)$ for $0 \leq z < 1$, with $\mu(\{0\}) = \bar{h}(0+)$, and with the remaining mass $1 - \bar{h}(1-)$ at 1. Conversely, for any such μ the function defined by the display is nondecreasing, right-continuous, at most G_0 and satisfies $H_L(\gamma)$, because it is a product of the nonnegative nondecreasing functions z^γ and $\mu([0, z])$; it tends to 0 at 0, so the law it describes has no mass at 0. The integral identity is Tonelli's theorem applied to $\mu([0, z]) = \int \mathbf{1}\{z_c \leq z\} \mu(dz_c)$. $\square$

The upper tail has the same structure: under $H_U(\eta)$, $\bar{G}(K_{\max}z) = \bar{G}_0 z^{-\eta} \nu((z, \infty])$ for $z > 1$ and a probability measure ν on $[1, \infty]$, and $\int_{K_{\max}}^\infty \bar{G}\phi du = \bar{G}_0 K_{\max} \int \int_1^{z_c} z^{-\eta} \phi(K_{\max}z) dz \nu(dz_c)$. For w integrable against P on $(0, K_{\min}]$, write $W(u) = \int_u^{K_{\min}} w$, so that $\int_0^{K_{\min}} wP dK = \int_0^{K_{\min}} GW du$ by Tonelli's theorem, and define on $[0, 1]$

$$a(z_c) = \frac{1 - z_c^{\gamma+1}}{\gamma + 1}, \quad b_w(z_c) = \int_{z_c}^1 z^\gamma W(K_{\min}z) dz = \frac{K_{\min}}{\gamma + 1} \int_{z_c}^1 w(K_{\min}z) (z^{\gamma+1} - z_c^{\gamma+1}) dz,$$

and on $[1, \infty]$ the upper-tail counterparts $a_U(z_c) = (1 - z_c^{1-\eta})/(\eta - 1)$ and $b_w^U(z_c) = \frac{K_{\max}}{\eta - 1} \int_1^{z_c} w(K_{\max}z) (z^{1-\eta} - z_c^{1-\eta}) dz$. By the lemma with $\phi = 1$, $P_0 = G_0 K_{\min} \int a d\mu$, and with $\phi = W$, $\int_0^{K_{\min}} wP dK = G_0 K_{\min} \int b_w d\mu$.

Proposition 24. (a) *Admissible exponents. Some law compatible with (P_0, G_0) satisfies $H_L(\gamma)$ if and only if $\gamma \leq \bar{\gamma} = K_{\min}G_0/P_0 - 1$, and some law compatible with $(C_0, \bar{G}_0)$ satisfies $H_U(\eta)$ if and only if $1 < \eta \leq \bar{\eta} = 1 + K_{\max}\bar{G}_0/C_0$. Here $\bar{\gamma} + 1$ is the strike elasticity of the put price at $K_{\min}$ and $\bar{\eta} - 1$ is minus that of the call price at $K_{\max}$. At $\gamma = \bar{\gamma}$ the lower tail is unique, $G(u) = G_0(u/K_{\min})^{\bar{\gamma}}$ on $(0, K_{\min}]$, so that $P(K) = P_0(K/K_{\min})^{\bar{\gamma}+1}$ there and every tail integral is point-identified; likewise $\bar{G}(u) = \bar{G}_0(u/K_{\max})^{1-\bar{\eta}}$ at $\eta = \bar{\eta}$. The hypotheses are nested: $H_L(\gamma)$ implies $H_L(\gamma')$ for $\gamma' < \gamma$, and $H_U(\eta)$ implies $H_U(\eta')$ for $1 < \eta' < \eta$.*

(b) *Sharp identified set. Let $0 < \gamma \leq \bar{\gamma}$, $1 < \eta \leq \bar{\eta}$ and let $w_1, \dots, w_m$ satisfy $\int_0^{K_{\min}} |w_j| K^{\gamma+1} dK < \infty$ and $\int_{K_{\max}}^\infty |w_j| K^{1-\eta} dK < \infty$. The set of vectors $(\int_0^{K_{\min}} w_j P dK)_{j \leq m}$ over the laws compatible with the observations and $H_L(\gamma)$ is*

$$S_L = G_0 K_{\min} \left\{ \left(\int b_{w_j} d\mu \right)_{j \leq m} : \mu \text{ a probability on } [0, 1], \int a d\mu = \frac{P_0}{G_0 K_{\min}} \right\},$$

a compact convex set each of whose extreme points is attained by a μ with at most two atoms; the upper tail gives S_U in the same way. The two tails vary independently, so the identified set of the raw moments (m_1, m_2, m_3) of $R3$ is the Minkowski sum $\text{mid} + S_L + S_U$, with mid the observed middle part. For $m = 1$, S_L is $G_0 K_{\min}$ times the interval between the lower convex and upper concave envelopes, at abscissa $P_0/(G_0 K_{\min})$, of the curve $z_c \mapsto (a(z_c), b_w(z_c))$.

(c) R4 revisited. If $w \geq 0$, the upper end of the identified interval of $\int_0^{K_{\min}} w P dK$ is $R4$'s bound $P_0 K_{\min}^{-(\gamma+1)} \int_0^{K_{\min}} w K^{\gamma+1} dK$, attained by a power law below $K_{\min}$ with an atom at $K_{\min}$, and the lower end is $G_0 K_{\min} b_w(z^*)$ with $a(z^*) = P_0/(G_0 K_{\min})$, attained by a law whose lower tail lies in $[K_{\min} z^*, K_{\min}]$. The lower end is positive if $w > 0$ on a set of positive measure in $(K_{\min} z^*, K_{\min})$, so $R4$'s upper bound is sharp and its lower bound of zero is not. The upper tail is analogous.

(d) Setting (i). Suppose G has a density g that is continuous and positive at $K_{\min}$, and let $\gamma_i = K_{\min} g(K_{\min})/G_0$. Then $H_L(\gamma)$ implies $\gamma \leq \gamma_i$, and $H_L(\gamma_i)$ holds if and only if the reversed hazard rate of $\ln X$, $ug(u)/G(u)$ at $x = \ln u$, attains its infimum over $(0, K_{\min}]$ at $K_{\min}$, which is the case if $\ln X$ has a log-concave distribution function. If $H_L(\gamma_i)$ holds, then $\gamma_i \leq \bar{\gamma}$. With $\varepsilon(K) = KP'(K)/P(K)$ and $\gamma_i(K), \bar{\gamma}(K)$ the same quantities at a general strike, $K\varepsilon'(K) = \varepsilon(K)(\gamma_i(K) - \bar{\gamma}(K))$, so $\gamma_i \leq \bar{\gamma}$ at $K_{\min}$ exactly when $\ln P$ is locally concave in $\ln K$ there. The same holds in the upper tail with $\eta_i = K_{\max} g(K_{\max})/\bar{G}_0$, the hazard rate of $\ln X$ and $\bar{\eta}$.

(e) Breakdown. For $\theta \in (0, 1]$ let $I(\theta)$ be the identified set of the variance and third central moment under $H_L(\theta\bar{\gamma})$ and $H_U(1 + \theta(\bar{\eta} - 1))$. Then $I(\theta') \subseteq I(\theta)$ for $\theta < \theta'$, and $I(1)$ is a single point. The set of θ at which the sign of the third central moment, and hence of the skewness, is identified is therefore empty or an interval with right end 1, and its infimum θ^* is the breakdown point of the sign.

Proof. (a) By the lemma, $P_0/(G_0 K_{\min}) = \int a d\mu$, and a is continuous and strictly decreasing from $a(0) = 1/(\gamma + 1)$ to $a(1) = 0$. A feasible μ therefore exists if and only if $P_0/(G_0 K_{\min}) \leq 1/(\gamma + 1)$, that is $\gamma \leq \bar{\gamma}$; for such γ the mixture of δ_0 and δ_1 with weight $(\gamma + 1)P_0/(G_0 K_{\min})$ on δ_0 is feasible. At $\gamma = \bar{\gamma}$ the constraint reads $\int a d\mu = a(0)$, which forces $\mu = \delta_0$ because $a < a(0)$ on $(0, 1]$. The upper tail is the same with a_U , strictly increasing from 0 at $z_c = 1$ to $1/(\eta - 1)$ at $z_c = \infty$. The elasticity statements are the definitions, since $P'(K_{\min}+) = G_0$ and $C'(K_{\max}+) = -\bar{G}_0$. Nesting holds because $G/u^{\gamma'} = (G/u^{\gamma}) u^{\gamma-\gamma'}$ is a product of nonnegative nondecreasing functions, and similarly in the upper tail.

(b) By the lemma the map $\mu \mapsto \text{law}$ is a bijection between the probability measures on $[0, 1]$ satisfying the constraint and the laws below $K_{\min}$ compatible with (P_0, G_0) and $H_L(\gamma)$; each such law extends to a law on $(0, \infty)$ compatible with the observed prices, because the law on $(K_{\min}, K_{\max})$ is fixed by the smile and the mass and the mean F are fixed by P_0, G_0, C_0 and $\bar{G}_0$ (R2). The integrability conditions make a and b_{w_j} continuous on $[0, 1]$, so the feasible set of μ is weak* compact and convex and its image under the continuous linear map $\mu \mapsto (\int b_{w_j} d\mu)_j$ is compact and convex. Extreme points of the image are images of extreme points of the feasible set, which are mixtures of at most two Dirac measures (Winkler, 1988, Theorem 2.1, as quoted by Pinelis, 2012). For $m = 1$ the set of pairs $(\int a d\mu, \int b_w d\mu)$ over all probability measures is the convex hull of the compact curve $\{(a(z_c), b_w(z_c))\}$ (Kemperman, 1968), and its section at abscissa $P_0/(G_0 K_{\min})$ is the stated interval. The lower-tail law and the upper-tail law can be chosen separately, which gives the Minkowski sum.

(c) For $w \geq 0$, W is nonnegative and nonincreasing. Along the curve, $db_w/dz_c = -z_c^{\gamma} W(K_{\min} z_c)$ and $da/dz_c = -z_c^{\gamma}$, so b_w is a function of a with slope $W(K_{\min} z_c)$, which increases as a increases. The curve is therefore the graph of a convex function of a : its lower envelope is the curve itself, attained by the single atom z^* , and its upper envelope is the chord from $(0, 0)$ at $z_c = 1$ to $(a(0), b_w(0))$ at $z_c = 0$. At abscissa $P_0/(G_0 K_{\min})$ the chord gives $G_0 K_{\min} \cdot (\gamma + 1) \frac{P_0}{G_0 K_{\min}} b_w(0) = (\gamma + 1) P_0 \int_0^{K_{\min}} w(K) \frac{(K/K_{\min})^{\gamma+1}}{\gamma+1} dK$, which is $R4$'s bound. The

law is the mixture of δ_0 and δ_1 of the proof of (a). If $w > 0$ on a set of positive measure in $(K_{\min}z^*, K_{\min})$, then $b_w(z^*) > 0$.

(d) $\ln(G/u^\gamma)$ is nondecreasing on $(0, K_{\min}]$ and differentiable from the left at $K_{\min}$ with derivative $g/G - \gamma/u$, which is therefore nonnegative: $\gamma \leq \gamma_i$. With a density, $\ln G$ is locally absolutely continuous where $G > 0$, so $H_L(\gamma)$ holds if and only if $ug(u)/G(u) \geq \gamma$ almost everywhere on $(0, K_{\min}]$; with $\gamma = \gamma_i$ this says that the infimum is attained at $K_{\min}$. If $\ln X$ has a log-concave distribution function, its reversed hazard rate is nonincreasing and the infimum over $x \leq \ln K_{\min}$ is attained at the end point; a log-concave density suffices by Bagnoli and Bergstrom (2005, Theorem 1). The inequality $\gamma_i \leq \bar{\gamma}$ is (a) applied to γ_i . Finally, $\varepsilon = KG/P$ gives $K\varepsilon' = \varepsilon + \varepsilon Kg/G - \varepsilon^2 = \varepsilon(\gamma_i(K) + 1 - \varepsilon) = \varepsilon(\gamma_i(K) - \bar{\gamma}(K))$, and $K\varepsilon'(K)$ is the derivative of ε with respect to $\ln K$.

(e) By the nesting in (a), the laws compatible with the hypotheses at θ' are compatible with them at θ , so the identified sets are nested; at $\theta = 1$ both tails are unique by (a). If the sign is identified at θ , it is identified at every $\theta' > \theta$. $\square$

Remark 25. The implementation (`src/qef/fx/identification.py`) restricts μ and ν to a grid of step points graded towards the boundary, which gives an inner approximation that converges as the grid is refined; the ranges of the variance and of the third central moment are computed exactly for that grid by linear programmes in the measures with the mean fixed, and optimised over the mean. The weights of the moment contracts of R3 satisfy the integrability conditions of (b) for every $\gamma > 0$ and $\eta > 1$. Because $I(1)$ is a point, the Pareto tails of (a) give the unique moments at the strongest admissible hypothesis; the breakdown point measures how far below that hypothesis the sign of the skewness survives. In the model validation of `reports/model_validation.md`, the model's own tail exponents lie between 0.58 and 0.97 of the maximal ones, setting (i) fails in seven of eight cases, and the identified set at the model's own exponents contains the true moments in every case, as (b) requires.

An observed price beyond a boundary strike tests the hypotheses and, if consistent, adds a linear constraint on the mixing measure. The following consequence of the lemma is used for the out-of-sample test with 5Δ quotes.

Proposition 26. *Let $0 < K < K_{\min}$, $k = K/K_{\min}$ and $0 \leq \gamma \leq \bar{\gamma}$, where $\gamma = 0$ stands for the limit of the weakest hypothesis. Under $H_L(\gamma)$ the identified interval of $P(K)$ is*

$$\left[\max\left(0, P_0 - G_0 K_{\min} \frac{1 - k^{\gamma+1}}{\gamma + 1}\right), P_0 k^{\gamma+1} \right].$$

For $K > K_{\max}$, $k = K/K_{\max}$ and $1 \leq \eta \leq \bar{\eta}$, the identified interval of $C(K)$ under $H_U(\eta)$ is $[\max(0, C_0 - \bar{G}_0 K_{\max} a_k^U), C_0 k^{1-\eta}]$ with $a_k^U = (1 - k^{1-\eta})/(\eta - 1)$, read as $\ln k$ at $\eta = 1$. The upper ends decrease and the lower ends increase with the exponent, so an observed price at K is consistent with the hypothesis exactly for exponents in an interval that starts at the weakest hypothesis and may be empty.

Proof. By the lemma with $\phi = \mathbf{1}\{u \leq K\}$, $P(K) = \int_0^{K_{\min}} G \mathbf{1}\{u \leq K\} du = G_0 K_{\min} \int b d\mu$ with $b(z_c) = \int_{z_c}^1 z^\gamma \mathbf{1}\{z \leq k\} dz = \max(0, a(z_c) - a_k)$, where $a_k = (1 - k^{\gamma+1})/(\gamma + 1)$. As a function of a this is convex and piecewise linear, so the lower envelope at $P_0/(G_0 K_{\min})$ is the function itself, which gives the lower end, and the upper envelope is the chord from $(0, 0)$ to $(a(0), b(0)) = (1/(\gamma + 1), k^{\gamma+1}/(\gamma + 1))$, which gives $P_0 k^{\gamma+1}$, R4's bound. Since $a_k = \int_k^1 z^\gamma dz$ decreases in γ for $k < 1$, the lower end increases in γ , and $P_0 k^{\gamma+1}$ decreases. The upper tail is the same with $\bar{G}(K_{\max}z) = \bar{G}_0 z^{-\eta} \nu((z, \infty])$ and $b(z_c) = \int_k^{z_c} z^{-\eta} dz$ for $z_c \geq k$; $a_k^U = \int_1^k z^{-\eta} dz$ decreases in η for $k > 1$. $\square$

The Garman–Kohlhagen PDE

The equation, and the representation of its solution as an expectation, are standard (Garman and Kohlhagen, 1983; for the Feynman–Kac argument, Shreve, 2004). The verification and the uniqueness argument below are written out for completeness.

Let $dS_t = (r_d - r_f)S_t dt + \sigma S_t dW_t$ under $\mathbb{Q}^d$ with constant coefficients, and let $c(t, S)$ be the Garman and Kohlhagen (1983) call price with strike K and expiry T ,

$$c(t, S) = Se^{-r_f(T-t)}\Phi(d_+) - Ke^{-r_d(T-t)}\Phi(d_-), \quad d_{\pm} = \frac{\ln(S/K) + (r_d - r_f \pm \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}.$$

Proposition 27. (a) $c \in C^{1,2}([0, T] \times (0, \infty))$ satisfies

$$\partial_t c + (r_d - r_f)S \partial_S c + \frac{1}{2}\sigma^2 S^2 \partial_{SS} c - r_d c = 0,$$

$0 \leq c \leq Se^{-r_f(T-t)}$, and $c(t, S) \rightarrow (S - K)^+$ as $t \uparrow T$, locally uniformly in S . (b) If $u \in C^{1,2}([0, T] \times (0, \infty)) \cap C([0, T] \times (0, \infty))$ solves the same equation with $u(T, S) = (S - K)^+$ and $|u(t, S)| \leq a(1 + S^m)$ for some a, m , then $u = c$. No boundary condition at $S = 0$ is required.

Proof. (a) Write $\tau = T - t$. The identity $Se^{-r_f\tau}\varphi(d_+) = Ke^{-r_d\tau}\varphi(d_-)$ gives $\partial_S c = e^{-r_f\tau}\Phi(d_+)$ and $\partial_{SS} c = e^{-r_f\tau}\varphi(d_+)/(S\sigma\sqrt{\tau})$. The same identity gives $\partial_t c = r_f Se^{-r_f\tau}\Phi(d_+) - r_d Ke^{-r_d\tau}\Phi(d_-) - Se^{-r_f\tau}\varphi(d_+)\sigma/(2\sqrt{\tau})$. Substitution verifies the equation. The bounds follow from $c = e^{-r_d\tau}\mathbb{E}[(S_T - K)^+ | S_t = S]$ and $\mathbb{E}[S_T | S_t = S] = Se^{(r_d - r_f)\tau}$. As $\tau \downarrow 0$, $\Phi(d_{\pm}) \rightarrow \mathbf{1}\{S > K\}$ for $S \neq K$ and $c(t, K) \rightarrow 0$, and c is Lipschitz in S with constant $e^{|r_f|T}$ uniformly in t , which gives locally uniform convergence.

(b) Fix (t, S) and let $\tau_n = \inf\{s \geq t : S_s \notin (1/n, n)\} \wedge (T - 1/n)$. Itô's formula for $e^{-r_d(s-t)}u(s, S_s)$ on $[t, \tau_n]$, the equation, and the boundedness of $\partial_S u$ on compact sets give $u(t, S) = \mathbb{E}[e^{-r_d(\tau_n-t)}u(\tau_n, S_{\tau_n})]$. Because S is a geometric Brownian motion started at $S > 0$, it reaches neither 0 nor ∞ before T , so $\tau_n \rightarrow T$ almost surely. Continuity of u on $[0, T] \times (0, \infty)$ then gives $u(\tau_n, S_{\tau_n}) \rightarrow (S_T - K)^+$. The growth bound and $\mathbb{E}\sup_{s \leq T} S_s^m < \infty$ allow dominated convergence, so $u(t, S) = e^{-r_d(T-t)}\mathbb{E}(S_T - K)^+ = c(t, S)$. The representation uses only values in the interior $(0, \infty)$, which the process never leaves, so no condition at $S = 0$ enters. Itô's formula and the Feynman–Kac representation are used in the forms given by Shreve (2004). $\square$

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